Exercise 10.1

1) No because BC = PQ
2) No because the angle must be an included angle
3)

(i) $\triangle APC \cong \triangle AQB$ because of the SAS rule. as AP = AQ, PC = QB and $\angle APC = \angle AQB$.

(ii) CP = BQ because it is a corresponding part of the congruent triangles

(iii) $\angle APC = \angle AQB$ because it is a corresponding part of the congruent triangles.

4) (i) As AB = AC and AP = AQ, the SAS rule can be applied over here.

(ii) ~~As both the sides of each triangle~~ have CP = BQ because it is a corresponding part of the congruent triangles.

(iii) $\angle ACP = \angle ABQ$ because it is a corresponding part of the congruent triangles.

5) Given that;
→ AD = BC
→ BD = AC

Using the SSS rule we know that $\triangle ABD = \triangle BAC$.

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(i) As the two triangles are congruent, the two angles have to be equal as the corresponding parts of congruent triangles.

6) Given that,

~~(i) AD = BC~~

→ AD = BC

→ ∠DAB = ∠CBA

(i) Here we can use the SAS rule. as AB is shared, BC = AD and ∠DAB = ∠CBA, the two triangles are congruent,

(ii) As AC and BD are corresponding sides of two congruent triangles, AC = BD.

(iii) As ∠ABD and ∠BAC are corresponding angles of two ~~triangles~~ congruent triangles, they are equal.

7) Given that:

→ AB = DC

→ AB ∥ DC

As AB and DC are parallel lines, ∠ABD = ∠CDB.

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As the two triangles share a common side (BD), AB = DC and ∠ABD = ∠CDB we use the SAS rule and prove that △ABD = △CDB.

8) Given that,
→ AC = AE
→ AB = AD
→ ∠BAD = ∠CAE

As AC = AE, AB = AD and ∠BAD = ∠CAE, Using the SAS rule, we can say that △ABC is congruent to △ADE.

9) As AB = CD we can say that AB + BC = CD + BC ⇒ AC = BD.

(i) As CE = BF, AC = BD and ∠ACE = ∠DBF, by applying the SAS rule we can prove that

△ACE ≅ △DBF

(ii) As AE and DF are corresponding sides of two congruent triangles, we can prove that AE = DF.

10) (i) As D is the midpoint of BC, we can say that BD = DC.

So as BD = DC, AB = AC and AD = AD, we use the SSS rule to say that △ADB ≅ △ADC.

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(ii) AD is a bisector of ∠A because it splits ~~ABC~~ ∠A into two equal halves and they are ∠BAD and ∠DAC.

(iii)

11) As AO = BO, CO = DO and ∠AOC = ∠BOD (vertically opposite angles), using the SAS rule we can say that ΔAOC ≅ ΔBOD.

(i) As AC and BD are corresponding parts of two congruent triangles, we can say that AC = BD.

(ii) As ~~AOD~~ AO = BO, DO = CO and ∠COB = ∠DOB (vertically opposite angles), using the SAS rule we can say that ΔAOD ≅ ΔCOB.

As ∠CAB and ∠ABD are corresponding parts of two congruent triangles, we can say that they are equal.

(iii)

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